SVM

1. **Setup Environment**
2. **Load Dataset**
    a. Select a dataset (any suitable classification dataset).
    b. Identify features and target variable.
3. **Preprocess Data**
    a. Handle missing values if any.
    b. Normalize or standardize features (important for SVM performance).
    c. Split dataset into training and testing sets.
4. **Initialize SVM Classifier**
    a. Choose kernel type (linear, polynomial, RBF, sigmoid (at least two)).
    b. Set parameters like C (regularization) and gamma (for non-linear kernels).
5. **Train the Model**
    a. Fit the SVM classifier on the training dataset.
6. **Make Predictions**
    a. Use the trained SVM model to predict outcomes on the test dataset.
7. **Evaluate Model Performance**
    a. Calculate accuracy, precision, recall, and F1-score.
    b. Generate and analyze the confusion matrix.
8. **Visualize Results**
    a. For datasets with 2 features, plot decision boundaries.
    b. Compare margins and support vectors.
9. **Experiment with Kernels and Parameters**
    a. Try different kernels (linear, RBF, polynomial).
    b. Adjust hyperparameters (C, gamma) and observe changes in accuracy.
10. **Summarize Results**
    a. Report best-performing kernel and parameters.
    b. Highlight trade-offs between accuracy and complexity.